

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
compliance with U.S. and local country laws. By using this product
you
agree to comply with applicable laws and regulations. If you are
unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be
found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email
to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K
bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface f0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up

R1(config-if)#
```

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SW1

Physical Config CLI Attributes

IOS Command Line Interface

```
Hardware Board Revision Number  : 0x01

Switch Ports Model              SW Version            SW Image
------ ----- -----              ----------            ----------
*    1 26    WS-C2960-24TT-L    15.0(2)SE4            C2960-
LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version
15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by mnguyen


Press RETURN to get started!



Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

SW1(config)#interface vlan 1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state
to up

SW1(config-if)#ip default-gateway 10.10.10.2
SW1(config)#
```

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IOS Command Line Interface

```
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cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K
bytes of memory
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2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no
no

Press RETURN to get started!



Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface f0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0,
changed state to up
```

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SW1

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to up

SW1(config)#interface vlan 1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown

SW1(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state
to up

SW1(config-if)#ip default-gateway 10.10.10.2
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

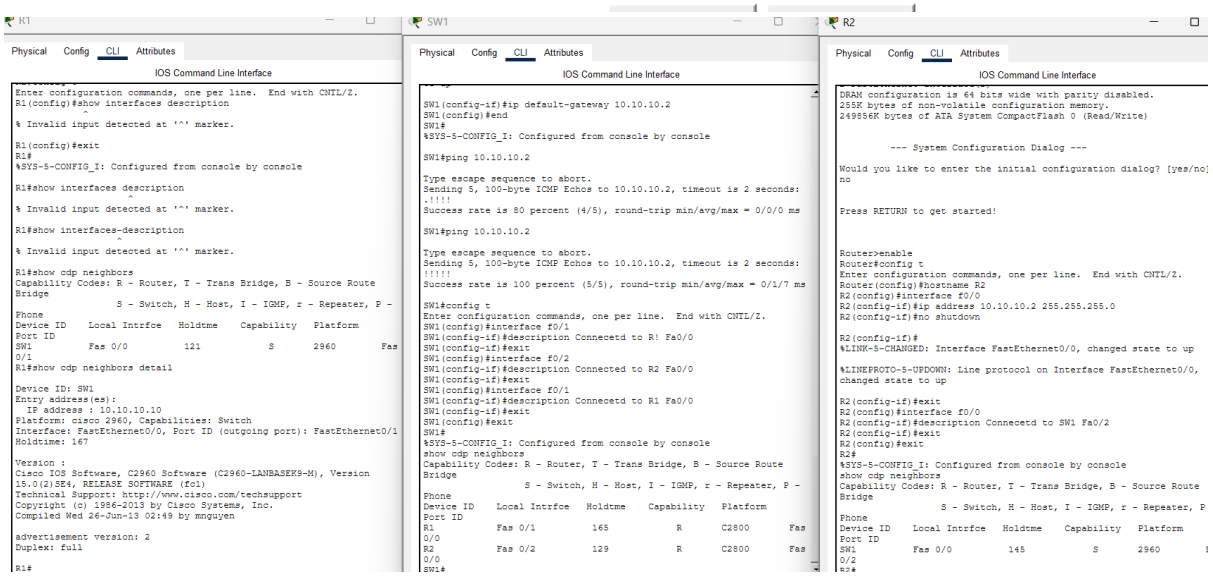
SW1#ping 10.10.10.2

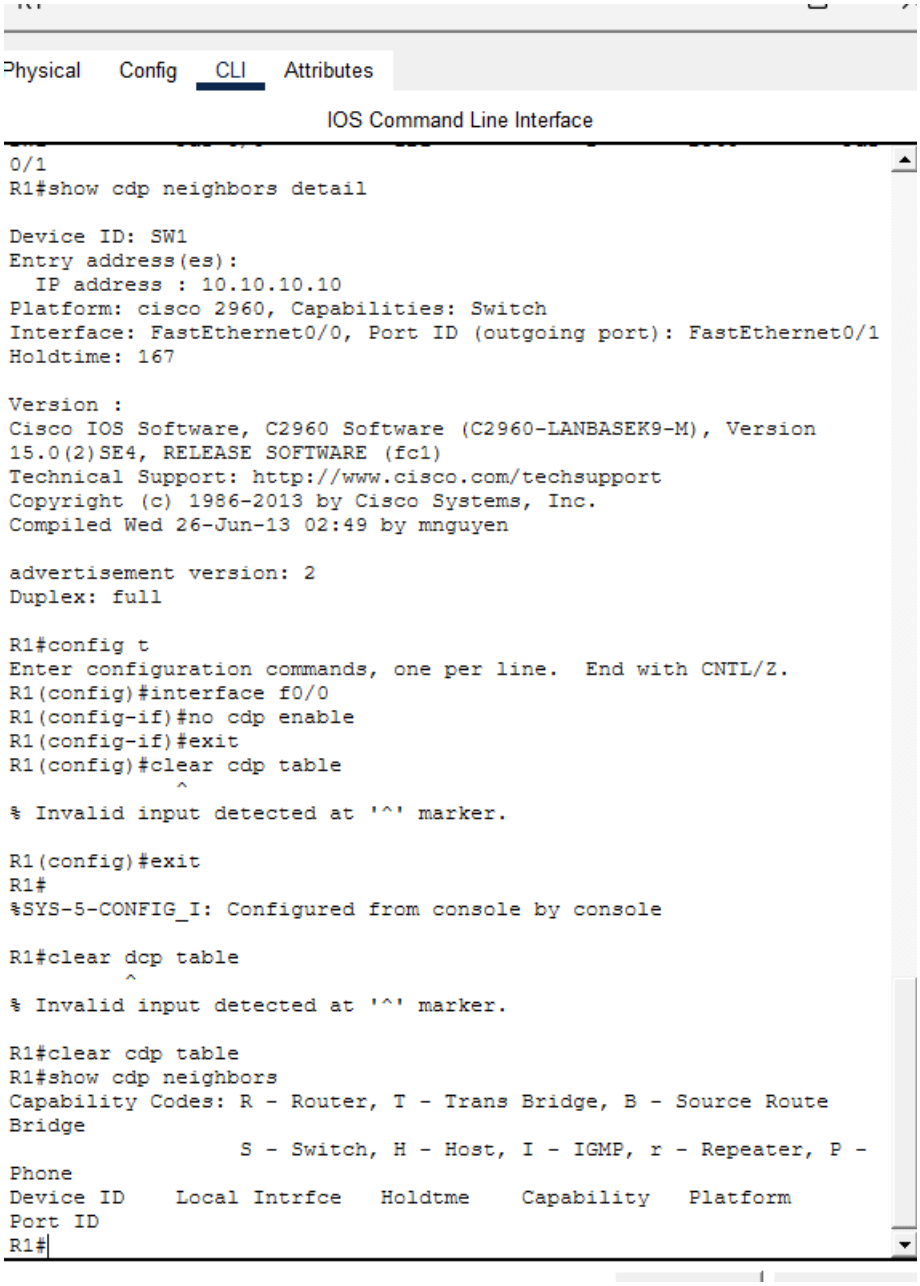
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

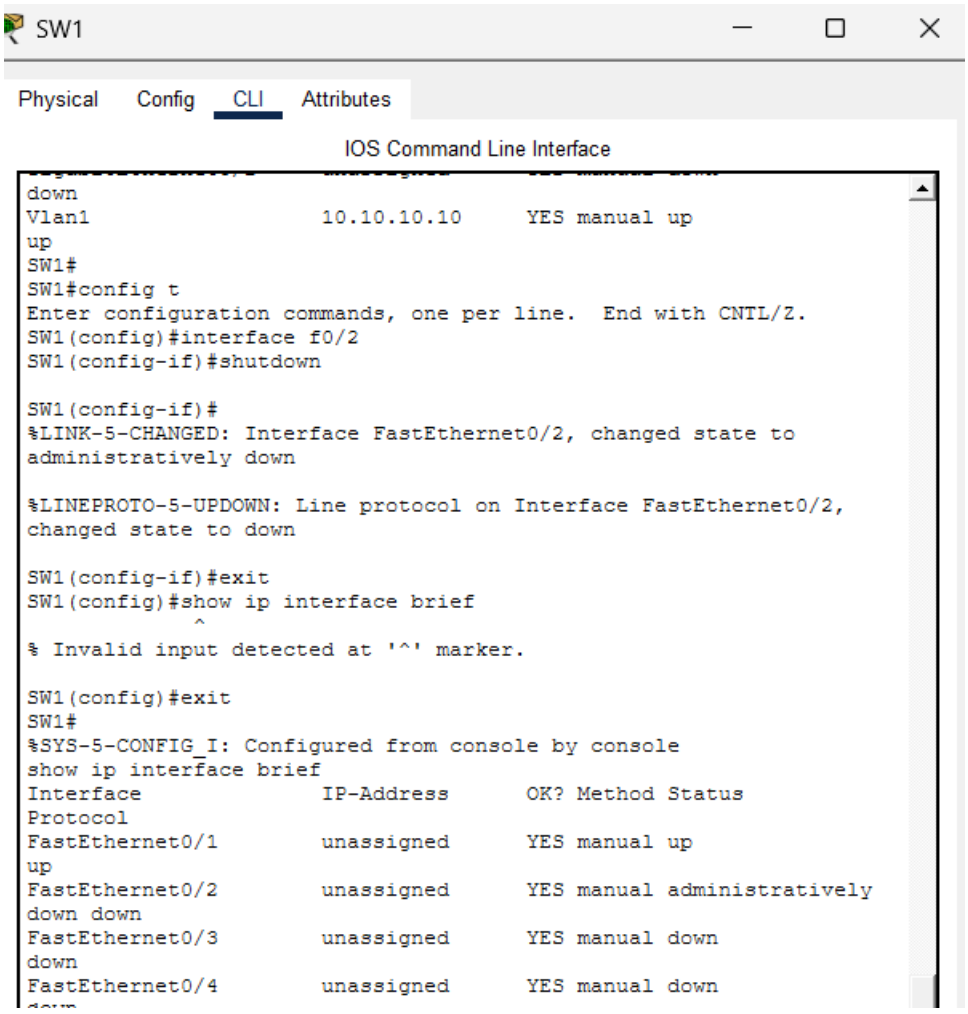
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/1/7 ms

SW1#
```




Physical
Config
CLI
Attributes
IOS Command Line Interface
0/1
R1#show cdp neighbors detail
Device ID: SW1
Entry address(es):
IP address : 10.10.10.10
Platform: cisco 2960, Capabilities: Switch
Interface: FastEthernet0/0, Port ID (outgoing port): FastEthernet0/1
Holdtime: 167
Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version
15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen
advertisement version: 2
Duplex: full
R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface f0/0
R1(config-if)#no cdp enable
R1(config-if)#exit
R1(config)#clear cdp table
^
% Invalid input detected at '^' marker.
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
R1#clear dcp table
^
% Invalid input detected at '^' marker.
R1#clear cdp table
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route
Bridge
S - Switch, H - Host, I - IGMP, r - Repeater, P -
Phone
Device ID Local Intrfce Holdtme Capability Platform
Port ID
R1#


SW1
Physical
Config
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IOS Command Line Interface
down
Vlan1 10.10.10.10 YES manual up
up
SW1#
SW1#config t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#interface f0/2
SW1(config-if)#shutdown
SW1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to
administratively down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2,
changed state to down
SW1(config-if)#exit
SW1(config)#show ip interface brief
^
% Invalid input detected at '^' marker.
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface IP-Address OK? Method Status
Protocol
FastEthernet0/1 unassigned YES manual up
up
FastEthernet0/2 unassigned YES manual administratively
down down
FastEthernet0/3 unassigned YES manual down
down
FastEthernet0/4 unassigned YES manual down

```
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces f0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0009.7c2b.bb02 (bia 0009.7c2b.bb02)
  Description: Connected to R2 Fa0/0
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
     2357 packets output, 263570 bytes, 0 underruns
     0 output errors, 0 collisions, 10 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out

SW1#
SW1#
SW1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface f0/2
SW1(config-if)#duplex half
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces f0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0009.7c2b.bb02 (bia 0009.7c2b.bb02)
  Description: Connected to R2 Fa0/0
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
```

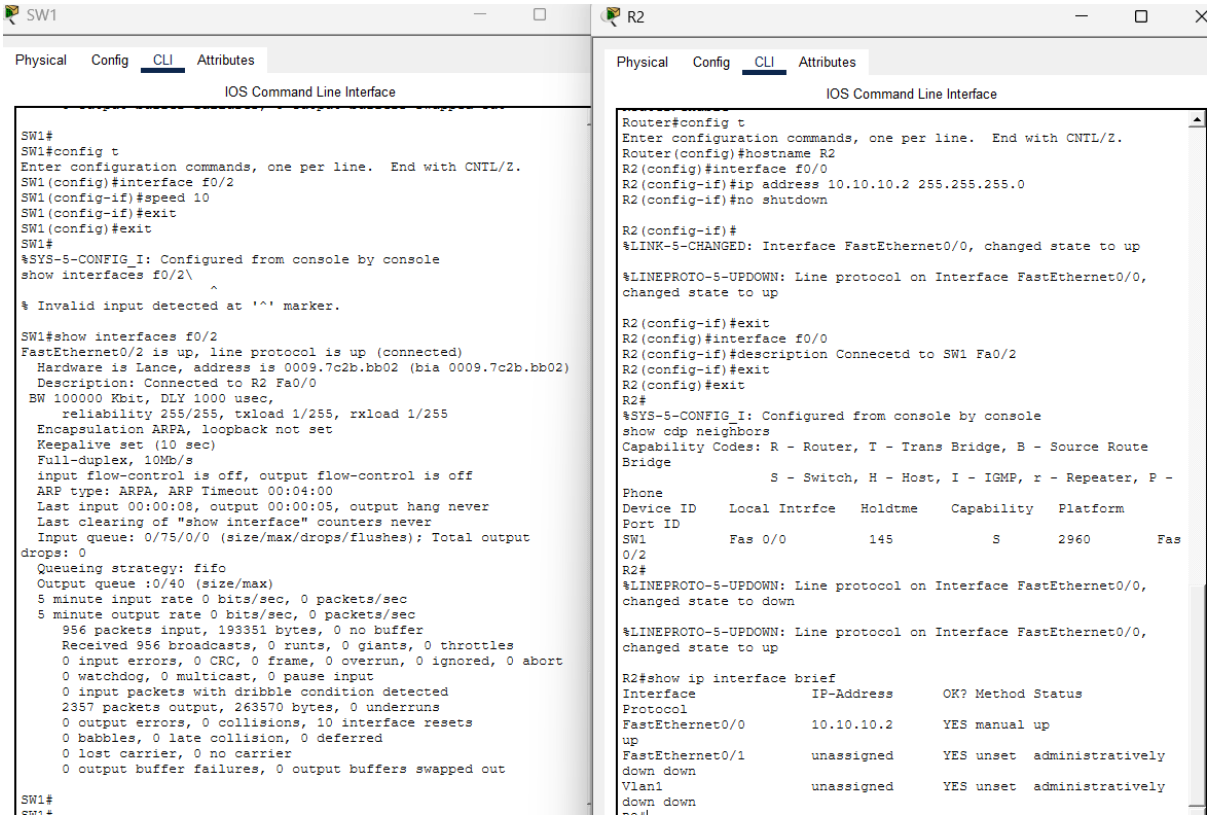
SW1

Physical Config CLI Attributes

```
  Description: Connected to R2 Fa0/0
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
     Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 watchdog, 0 multicast, 0 pause input
     0 input packets with dribble condition detected
     2357 packets output, 263570 bytes, 0 underruns
     0 output errors, 0 collisions, 10 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out

SW1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#interface f0/2
SW1(config-if)#duplex full
SW1(config-if)#exit
SW1(config)#exit
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show interfaces f0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0009.7c2b.bb02 (bia 0009.7c2b.bb02)
  Description: Connected to R2 Fa0/0
 BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     956 packets input, 193351 bytes, 0 no buffer
```



When a duplex mismatch occurs between SW1 and R2, network performance can decrease significantly. The connection may remain physically up, but communication becomes unreliable. Symptoms can include slow network performance, packet loss, retransmissions, collisions and late collisions.

This happens because one device operates in half-duplex mode while the other operates in full-duplex mode. Half-duplex devices expect to transmit or receive data at different times, whereas full-duplex devices can transmit and receive simultaneously. This difference causes communication problems because the devices do not handle data transmission in the same way.

As a result, packets may be lost or corrupted, requiring retransmission. This reduces the overall efficiency and performance of the network.